

Maven Electronics — Business Insights & Recommendations

1. Project Context

Maven Electronics is a global electronics retailer selling products through online and physical-store channels. As Data Analyst, the objective was to consolidate the provided sales data into an exploratory analysis that management could use to understand business performance and investigate the company's revenue decline following the 2019 peak.

The analysis examined revenue, orders, customers, units sold, average order value (AOV), products, categories, countries, sales channels, and operational performance.

2. Business Insights & Recommendations

2.1 Revenue Fell Mainly Because There Were Fewer Orders

Revenue fell by 49.1% from 2019 to 2020.

At the same time:

- Orders fell from 9,083 to 4,635 (-49.0%)
- Units sold fell from 68,440 to 34,463 (-49.6%)
- AOV only fell from \$2,010.83 to \$2,005.31 (-0.3%)

Customers were not spending much less each time they placed an order. Instead, the company simply had far fewer orders and fewer units being sold.

Recommendation

Maven should focus on bringing back customers, increasing repeat purchases, and generating more orders rather than focusing mainly on increasing AOV.

2.2 The Number of Active Customers Also Fell

Active customers decreased from 6,497 in 2019 to 3,868 in 2020, a 40.5% decrease.

Orders per customer also fell from 1.40 to 1.20.

So, there were fewer customers buying from Maven, and the customers who remained were also placing fewer orders.

Recommendation

Maven should focus on customer retention and bringing inactive customers back. It should also find ways to encourage existing customers to purchase more frequently.

2.3 The Decline Happened Across the Business

The decline was not limited to one area.

- Every country experienced a revenue decline.
- Every product category declined.
- Both Online and Physical sales declined.

So, the problem was not simply one bad country, product, or sales channel. The business experienced a broad decline in sales activity.

Recommendation

Maven should look at the problem as a company-wide sales and customer activity issue, rather than trying to fix only one market or product.

2.4 The United States Had the Biggest Dollar Loss

The United States lost approximately \$3.69M in revenue between 2019 and 2020.

Germany had one of the largest percentage declines, at approximately 59%, but the US had a much larger overall revenue base.

So, Germany had a bigger decline relative to its size, but the US had the biggest impact on the company's total revenue because it was a much larger market.

Recommendation

Maven should pay particular attention to the US market because recovering even part of its lost revenue could have a large impact on total company performance.

At the same time, smaller markets with large percentage declines should also be investigated.

2.5 Computers Were the Biggest Contributor to Revenue Loss

Computers contributed the largest revenue loss in dollar terms.

However, Home Appliances and Audio experienced some of the largest percentage declines.

We can say that computers had the biggest impact because they are such an important part of Maven's business.

But categories such as Home Appliances and Audio also need attention because their performance dropped sharply.

Recommendation

Maven should protect its strong Computer business while investigating why categories such as Home Appliances and Audio declined so heavily.

2.6 Revenue Depends Heavily on Certain Areas

Some areas contribute much more revenue than others:

- In-Store: \$44M
- Online: \$11M
- Computers: \$19.3M
- North America: \$27.4M

These areas are important strengths for Maven, but the company also depends heavily on them.

Recommendation

Maven should continue supporting these strong areas while also trying to grow online sales, develop other product categories, and expand attractive markets.

2.7 Delivery Performance Improved

Average delivery time improved significantly:

Year	Average Delivery Time
2016	7.3 days
2019	4.1 days
2020	4.0 days
2021	3.8 days

The overall average delivery time was 4.53 days, so this means delivery actually became faster over time.

The revenue decline after 2019 does not appear to be because Maven's delivery performance became worse.

Recommendation

Maven should continue maintaining its improved delivery performance, but its main focus should be on customers, orders, and sales demand.

3. Overall Conclusion

The biggest finding from the analysis is:

Maven Electronics' revenue decline in 2020 was mainly caused by fewer customers, orders, and units sold, rather than customers spending less per order. The decline happened across countries, sales channels, and product categories, so it was a company wide issue. Maven should focus on bringing back inactive customers, increasing repeat purchases and order volume, while protecting its strongest products and markets. The company should also reduce

its dependence on in store sales, Computers, and North America by growing other channels and markets, while continuing to improve delivery performance.

At the same time, Maven's delivery performance improved, which means the company should focus more on recovering sales and customer activity rather than trying to solve a delivery problem.

Recommended Priorities

1. Bring back inactive customers
2. Increase repeat purchases and order frequency
3. Recover lost order volume
4. Investigate why all major markets and categories declined
5. Protect the strong Computers category and US market
6. Grow the Online channel
7. Continue improving delivery performance